Leonid Nekrylov

Robot Learning Engineer

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Education

Moscow Institute of Physics and Technology (MIPT) *2023 – 2027*
BSc, Phystech School of Applied Mathematics and Computer Science Dolgoprudny, Russia
Major: Applied Mathematics and Physics

Relevant courses

- **Optimization and ML:** Year-long course on optimization methods in ML (D. Merkulov), Machine Learning 1 (R. Neychev), LLMs (Giraffe AI)
- **Reinforcement Learning:** Self-studied UC Berkeley CS285 (*Deep Reinforcement Learning*, Sergey Levine).
- **Mathematics:** Probability Theory, Stochastic Processes, Mathematical Statistics, Linear Algebra, Calculus and Functional Analysis.
- **Computer Science:** Algorithms and Data Structures, C++, Databases (PostgreSQL), Python.
- **Physics/Engineering:** Introduction to Robotics, Theoretical Mechanics.

Experience and projects

Pet or Threat, behaviour of a Unitree Go1 robot dog *October 2026*
Personal project, MuJoCo *Moscow, Russia*

▶ Watch the video leonidnn.github.io/pet-or-threat-site

- A simulated robot dog that nobody gives commands to. It decides on its own what to do, for example guard, play, cuddle, wait for its owner or sleep.
- The key architectural idea is that the decision module receives only perception and emits only intents such as go there or look here, while walking is handled by pretrained RL policies. So the module does not depend on the simulator and moves to another robot by replacing the lower level.

MIPT CCM, RL agent for RGB-D object grasping *September 2026 – October 2026*
Research project, Isaac Sim, UR10e *Moscow, Russia*

▶ Watch the video and code github.com/leonidnn/isaac-rgbd-grasping

- Built an Isaac Sim environment for grasp training with object randomization. Wrote a custom Vulkan layer that keeps the renderer off other people's GPUs on the shared server.
- Trained an RL agent. A network predicts grasp success for every point and angle from a depth map, and the agent takes the best one.
- Evaluation in new random poses with the full arm approach gave 18/20 and 19/20 on two objects. SAC did not learn at all in 5.7k attempts.
- Added a new object, a bag of chips, and compared fine-tuning approaches that avoid forgetting the old ones. The best so far is two networks with a switch by object type, giving 88% on the new object versus 42% without fine-tuning while keeping 95% on the old ones.
- Found a rendering artefact that had distorted training, and fixed the setup.

NB-Technologies, Hybrid Solver *September 2025 – November 2025*
LLM developer *Moscow, Russia*

▶ Try the solver huggingface.co/spaces/leonid-nn/hybrid-solver

▶ See the code github.com/leonidnn/hybrid-solver

- A math problem solver that either gives the correct answer or honestly says it could not get one.
- The key idea is to turn a language model from a Monte Carlo algorithm, which always answers but is sometimes wrong, into a Las Vegas algorithm, as in the ZPP class. Three independent methods solve the problem, model reasoning, exact computation in SymPy and a numerical check, and an answer is given only if all of them agree.

Cool Deal, wearable climate-control systems *December 2024 – September 2025*
Founder, project lead *Moscow, Russia*

- Led a team of 5, three engineers, a sales lead and myself, from idea to a working prototype of a wearable climate-control system.

- Designed and built prototypes: CAD, sensors, microcontroller firmware, bench testing.
- Raised a 1M rouble “Student Startup” grant for R&D and testing.
- Handled sales. Ran 2 pilot deployments, tested the hypothesis based on their results and closed the project.

Sber (Recommender Systems Team / Amazme) *July 2024 – November 2024*
Data Scientist *Moscow, Russia*

- Built PySpark data marts from raw clickstream.
- Fine-tuned candidate-retrieval models in PyTorch and evaluated them on offline metrics.
- Handled bug reports and traced the causes of anomalies.
- Wrote unit tests.

Technical skills

- Python (and libraries), PyTorch, PostgreSQL, Git, Pytest, Linux, LaTeX, Docker, NVIDIA Isaac Sim / Isaac Lab, MuJoCo, ROS2; C++ at code-reading level
- Robotics: forward and inverse kinematics (FK/IK), manipulator Jacobian, singularities and manipulability.
- English – C1 (TOEFL iBT 107). Russian – native.